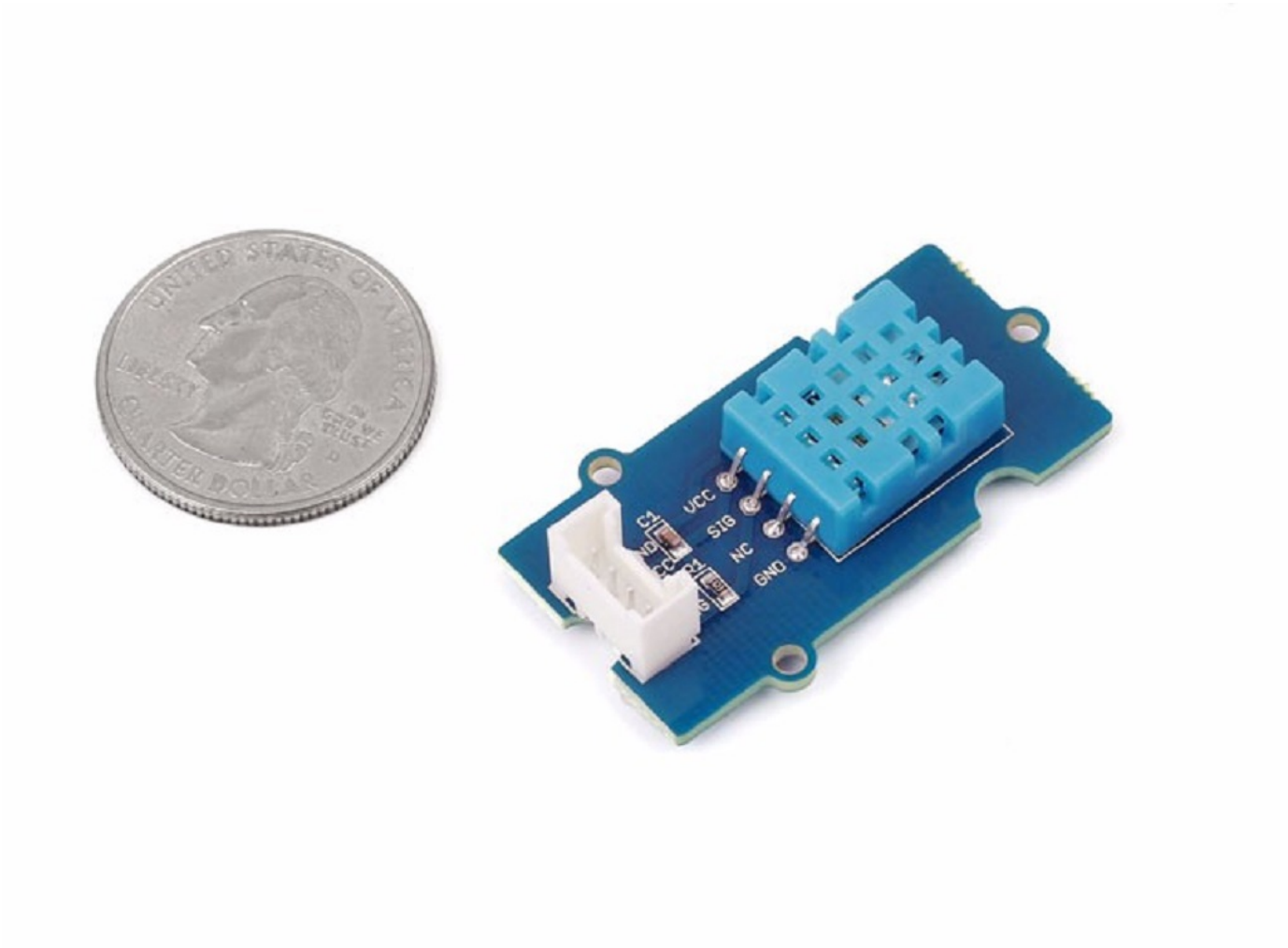
Grove - Temperature&Humidity Sensor (DHT11)



This Temperature&Humidity sensor provides a pre-calibrated digital output. A unique capacitive sensor element measures relative humidity and the temperature is measured by a negative temperature coefficient (NTC) thermistor. It has excellent reliability and long term stability. Please note that this sensor will not work for temperatures below 0 degree.

Get One Now

Upgradable to Industrial Sensors

With the SenseCAP S2110 controller and S2100 data logger, you can easily turn the Grove into a LoRaWAN® sensor. Seeed not only helps you with prototyping but also offers you the possibility to expand your project with the SenseCAP series of robust industrial sensors.

SenseCAP S210x series industrial sensors provide an out-of-box experience for environmental sensing. Please refer to the S2101 Wireless Temperature and Humidity Sensor with higher performance and robustness for air quality monitoring. The series includes sensors for soil moisture, air temperature and humidity, light intensity, CO2, EC, and an 8-in-1 weather station. Try the latest SenseCAP S210x for your next successful industrial project.

SenseCAP Industrial Sensor
S2101 Air Temp & Humidity

Features

- Relative Humidity and temperature measurement
- Full range temperature compensation Calibrated
- Digital signal
- Long term stability
- Long transmission distance(>20m)
- Low power consumption

> **TIP**
>
> More details about Grove modules please refer to Grove System

Applications Ideas

- Consumption product
- Weather station
- Humidity regulator
- Air conditioner

Specifications

Key Specifications

Items	Min
PCB Size	2.0cm*4.0cm
Interface	2.0mm pitch pin header
IO Structure	SIG,VCC,GND,NC
ROHS	YES

Electronic Characterstics

Items	Conditions	Min	Norm	Max	Unit
VCC	-	3.3	-	5	Volts
Measuring Current Supply	-	1.3	-	2.1	mA
Average Current Supply	-	0.5	-	1.1	mA
Measuring Range	Humidity	20%	-	90%	RH
	Temperature	0	-	50	°C

Accuracy	Humidity	-	-	±5%	RH
	Temperature			±2	°C
Sensitivity	Humidity		-	1%	RH
	Temperature			1	°C
Repeatability	Humidity			±1%	RH
	Temperature			±1	°C
Long-term Stability				±1%	RH/year
Signal Collecting Period			2		S

Platforms Supported

Arduino	Raspberry Pi			

NOTE

The platforms mentioned above as supported is/are an indication of the module's software or theoritical compatibility. We only provide software library or code examples for Arduino platform in most cases. It is not possible to provide software library / demo code for all possible MCU platforms. Hence, users have to write their own software library.

Getting Started